

Tense Timing Experiments: Analysis & Report

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Report

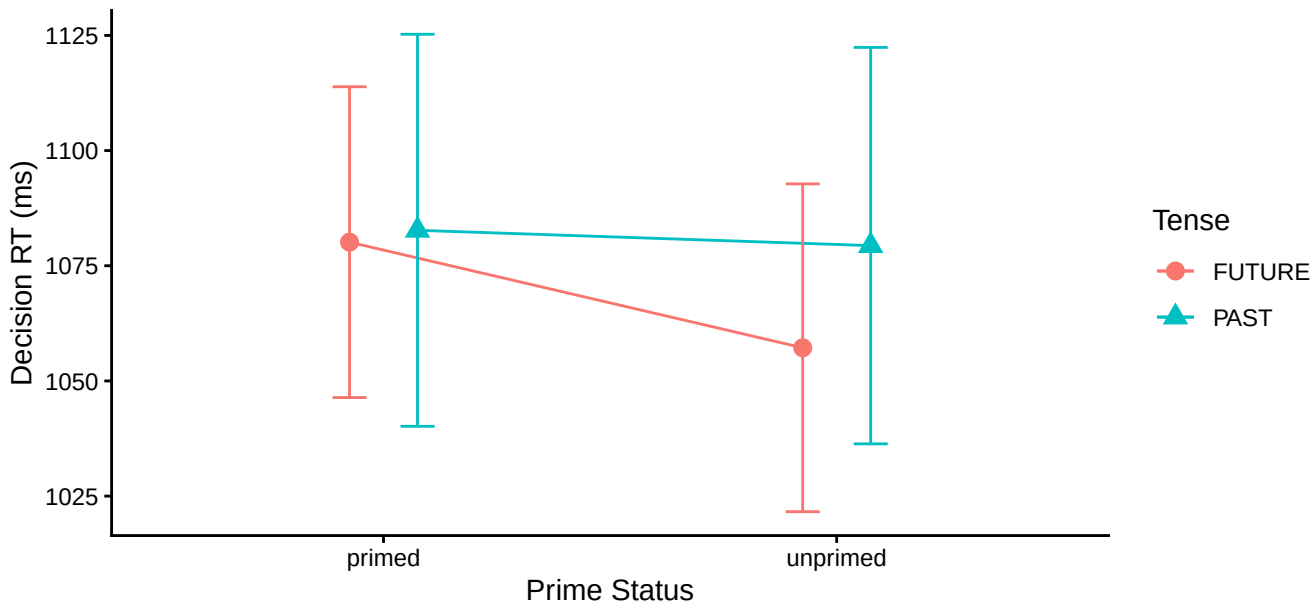
Experiment 1: Conceptual Task

24 participants learned character–verb–time pairings (Chef, Pirate, Wizard x 20 verbs) and made speeded keypress decisions (C = Past, M = Future). The key manipulation is **tense priming**: whether the current trial’s tense matches the previous trial’s (primed) or switches (unprimed). Out of 2592 total trials, 864 boundary trials (positions 1 and 12 of each 12-trial block) were excluded, leaving **1728 trials** in the analysis.

Table 1: Conceptual Task: Decision RT by Tense x Prime Status

Tense	Prime Status	N	Mean RT (ms)	SE
FUTURE	primed	432	1080	33.7
FUTURE	unprimed	432	1057	35.6
PAST	primed	432	1083	42.5
PAST	unprimed	432	1079	43.0

Conceptual Task: Decision RT



Error bars: ± 1 SE.

Experiment 2: Morphosyntax

3 participants produced sentences in past or future tense based on the same character–verb–time pairings (e.g., *The pirate spun a top* vs. *The pirate will spin a top*). Priming is by **tense repetition**, identical in logic to the conceptual task. Timing is measured from forced-aligned recordings.

Out of 300 total trials, 73 (24%) were excluded for incorrect tense and 100 were excluded as boundary trials (positions 1 and 12), leaving **151 trials** in the analysis.

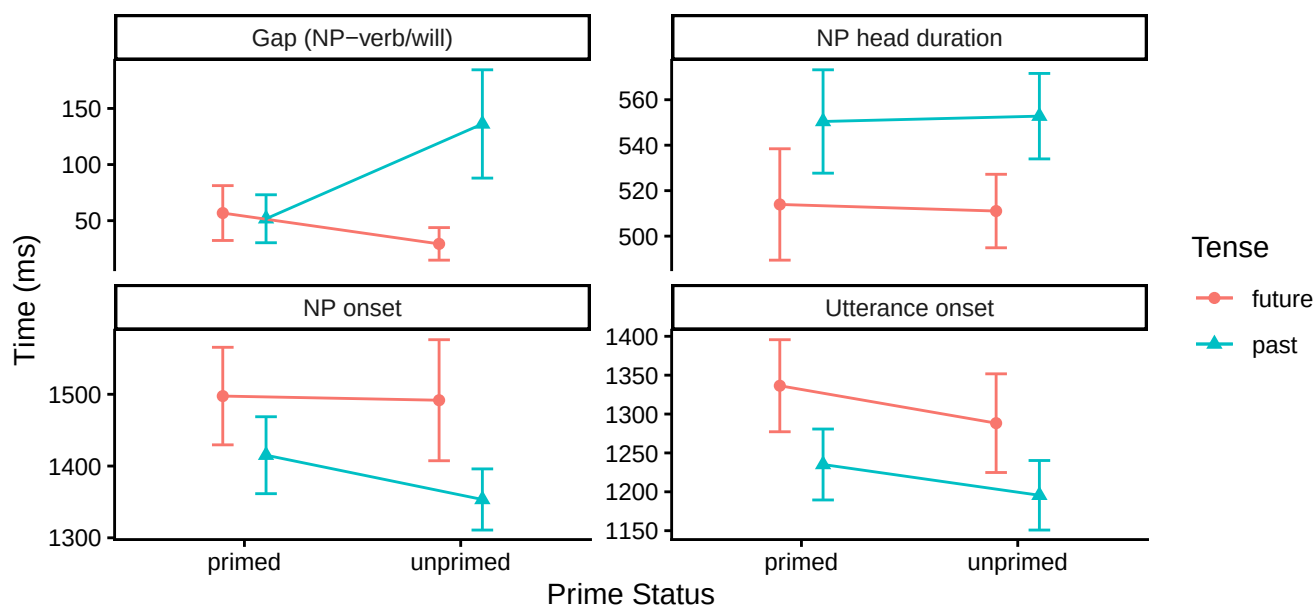
Measures:

- **Utterance onset** — when “the” begins (overall speech latency from trial start)
- **NP head duration** — duration of the entity name (e.g., *pirate*, *chef*, *wizard*)
- **Gap** — silence between NP offset and the critical word (*will* or past verb)
- **Critical word onset** — onset of *will* (future) or the past verb

Table 2: Morphosyntax: Mean timing measures by Tense x Prime Status (SE in parentheses)

Tense	Prime Status	N	Utt onset (ms)	NP dur (ms)	Gap (ms)	Crit onset (ms)
future	primed	28	1336.4 (59.2)	513.9 (24.5)	56.8 (24.4)	2068.2 (78.9)
future	unprimed	29	1288.3 (63.5)	511.0 (16.2)	29.3 (14.5)	2032.1 (95.1)
past	primed	47	1235.1 (45.6)	550.4 (22.7)	51.7 (21.4)	2017.2 (60.8)
past	unprimed	47	1195.5 (44.7)	552.8 (18.8)	136.2 (48.2)	2042.3 (76.9)

Morphosyntax: Timing Measures

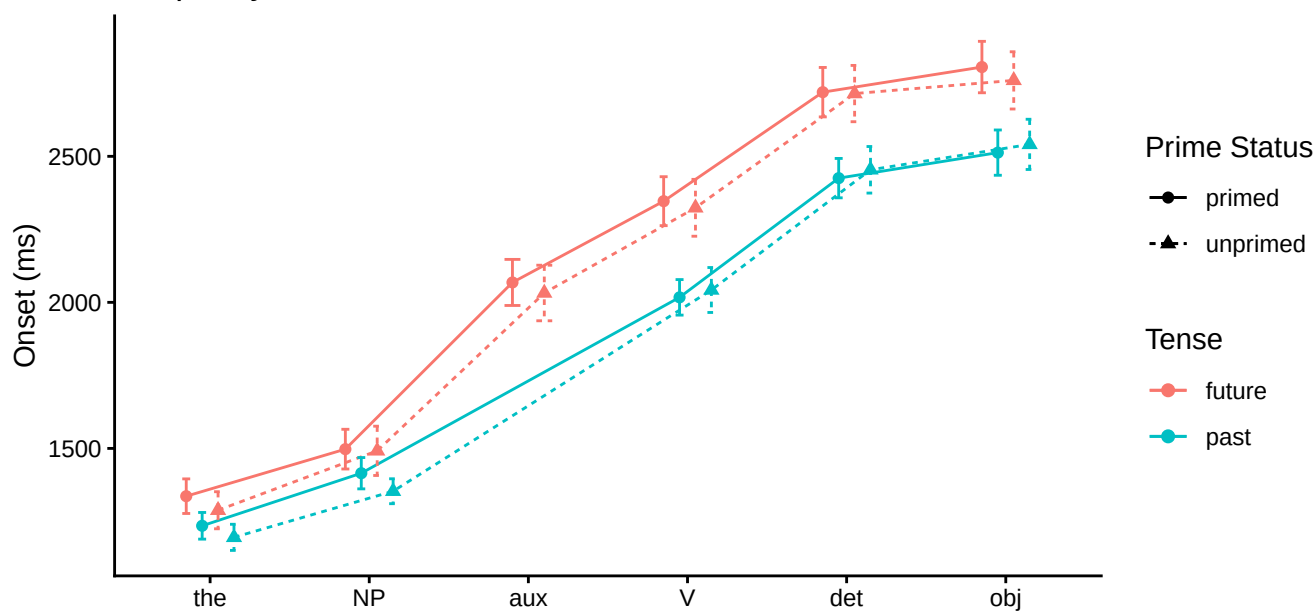


Error bars: ± 1 SE.

Word-by-Word Timeline

Onset

Morphosyntax: Word Onset



Error bars: ± 1 SE.

Table 3: Mean word onset (ms) by condition and word position (SE in parentheses)

condition	the	NP	aux	V	det	obj
future / primed	1336.4 (59.2)	1497.5 (68.0)	2068.2 (78.9)	2346.4 (83.8)	2719.6 (84.6)	2805.6 (88.0)
future / unprimed	1288.3 (63.5)	1491.7 (84.3)	2032.1 (95.1)	2323.8 (97.3)	2714.8 (96.3)	2760.0 (98.0)
past / primed	1235.1 (45.6)	1415.1 (53.6)	NA	2017.2 (60.8)	2425.3 (67.4)	2512.6 (77.5)
past / unprimed	1195.5 (44.7)	1353.4 (42.6)	NA	2042.3 (76.9)	2453.8 (79.7)	2540.9 (86.0)

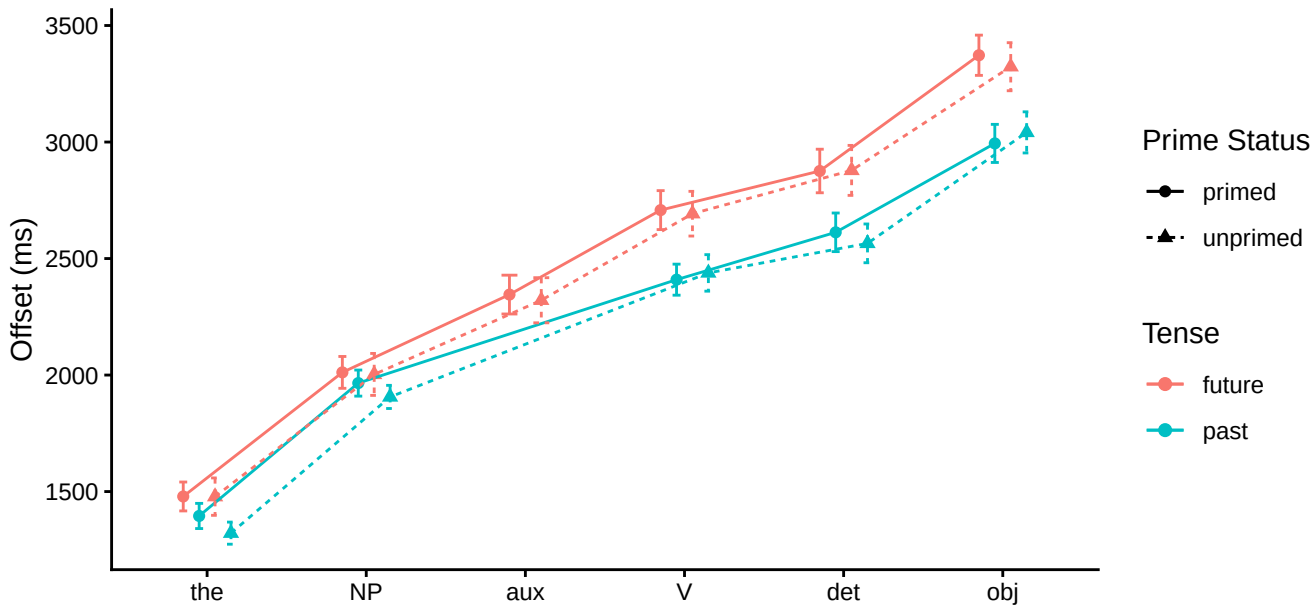
Duration

Table 4: Mean word duration (ms) by condition and word position (SE in parentheses)

condition	the	NP	aux	V	det	obj
future / primed	142.5 (12.8)	513.9 (24.5)	277.1 (13.3)	361.4 (17.6)	156.1 (35.6)	566.8 (15.5)
future / unprimed	190.0 (31.8)	511.0 (16.2)	289.0 (17.4)	368.6 (19.0)	163.4 (42.9)	563.2 (21.0)
past / primed	160.2 (19.4)	550.4 (22.7)	NA	392.1 (16.4)	187.2 (38.1)	481.4 (24.7)
past / unprimed	126.0 (9.8)	552.8 (18.8)	NA	396.0 (17.8)	111.3 (16.5)	500.4 (20.6)

Offset

Morphosyntax: Word Offset



Error bars: ± 1 SE.

Experiment 3: Morphophonology

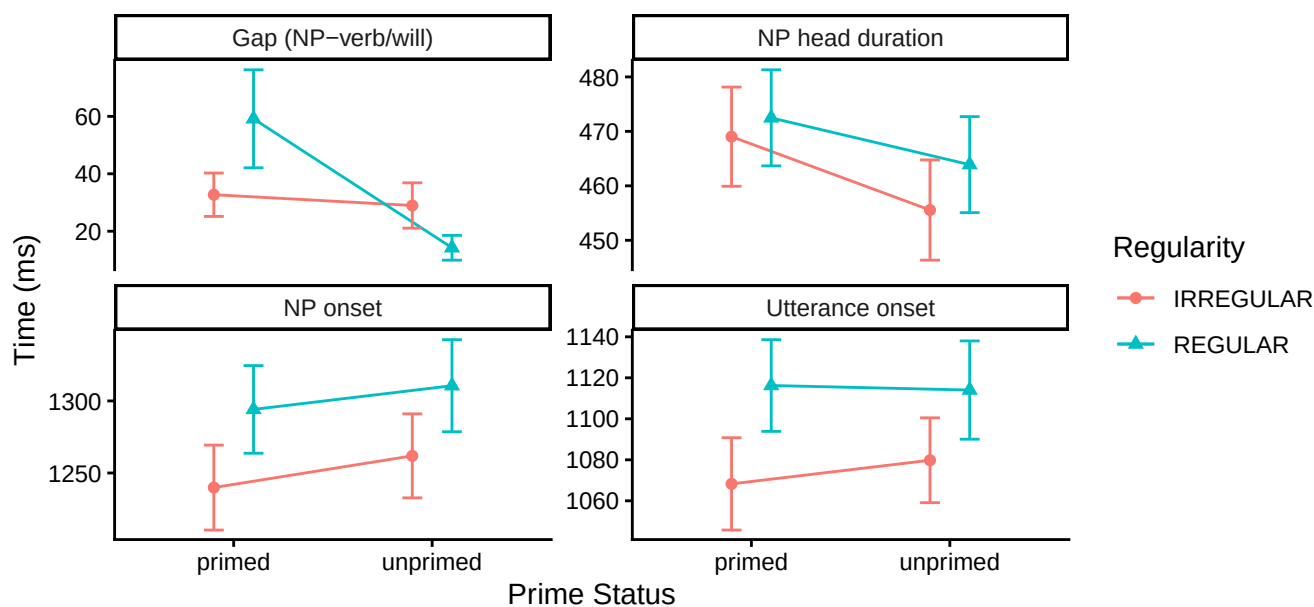
9 participants produced sentences in **past tense only**. The manipulation is **regularity priming**: whether the current verb's morphological class (regular: *-ed*, e.g., *washed*; irregular: suppletive, e.g., *drank*) matches the previous trial (primed) or switches (unprimed). Materials consist of 10 regular and 10 irregular verbs, frequency-matched.

Out of 960 total trials, 99 (10%) were excluded for incorrect tense and 320 were excluded as boundary trials, leaving **573 trials** in the analysis. Same timing measures as morphosyntax.

Table 5: Morphophonology: Mean timing measures by Regularity x Prime Status (SE in parentheses)

Regularity	Prime Status	N	Utt onset (ms)	NP dur (ms)	Gap (ms)	Crit onset (ms)
IRREGULAR	primed	144	1068.3 (22.5)	469.0 (9.1)	32.7 (7.6)	1741.6 (32.2)
IRREGULAR	unprimed	144	1079.8 (20.7)	455.6 (9.2)	29.0 (7.9)	1746.4 (32.7)
REGULAR	primed	141	1116.2 (22.4)	472.5 (8.8)	59.1 (17.1)	1825.7 (40.8)
REGULAR	unprimed	144	1114.0 (24.0)	463.9 (8.8)	14.2 (4.3)	1788.7 (34.5)

Morphophonology: Timing Measures

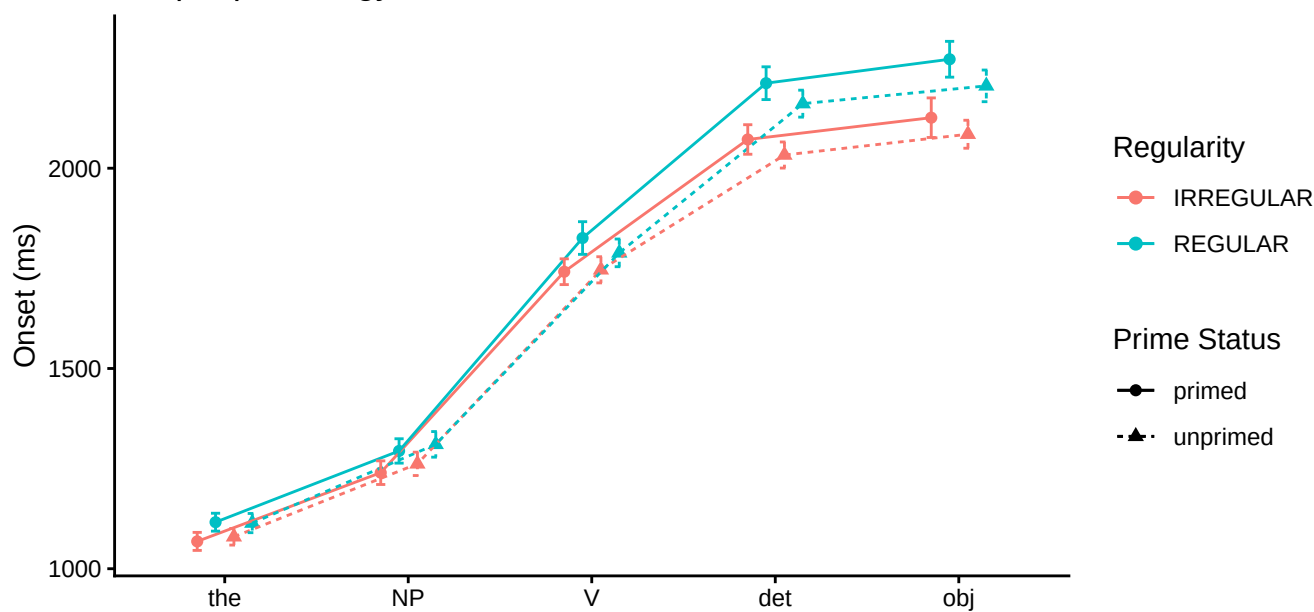


Error bars: ± 1 SE.

Word-by-Word Timeline

Onset

Morphophonology: Word Onset



Error bars: ± 1 SE.

Table 6: Mean word onset (ms) by condition and word position (SE in parentheses)

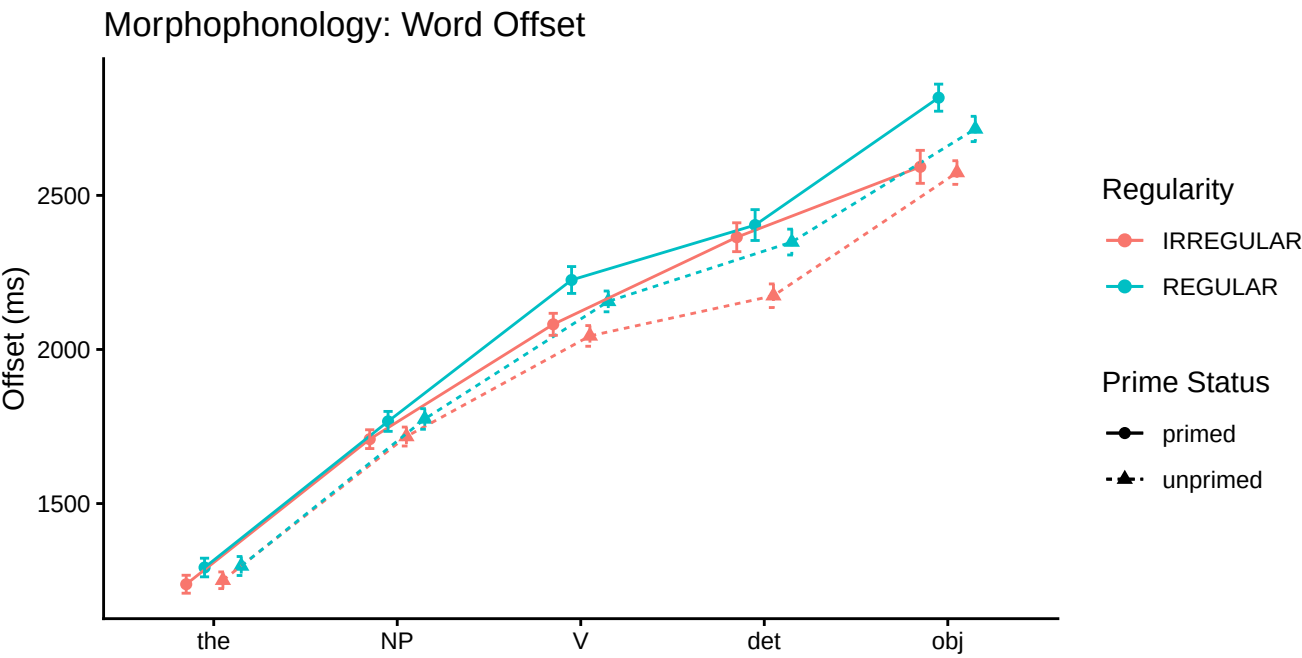
condition	the	NP	V	det	obj
IRREGULAR / primed	1068.3 (22.5)	1239.9 (29.5)	1741.6 (32.2)	2071.8 (36.9)	2126.2 (49.4)
IRREGULAR / unprimed	1079.8 (20.7)	1261.9 (29.1)	1746.4 (32.7)	2033.0 (32.5)	2084.9 (34.9)
REGULAR / primed	1116.2 (22.4)	1294.1 (30.4)	1825.7 (40.8)	2212.3 (40.9)	2272.1 (44.7)
REGULAR / unprimed	1114.0 (24.0)	1310.6 (32.0)	1788.7 (34.5)	2161.2 (33.7)	2205.7 (39.5)

Duration

Table 7: Mean word duration (ms) by condition and word position (SE in parentheses)

condition	the	NP	V	det	obj
IRREGULAR / primed	169.9 (13.7)	469.0 (9.1)	340.0 (13.3)	292.4 (26.2)	466.4 (11.6)
IRREGULAR / unprimed	171.2 (13.2)	455.6 (9.2)	297.6 (12.2)	141.3 (13.7)	489.3 (9.6)
REGULAR / primed	176.2 (14.4)	472.5 (8.8)	399.7 (12.9)	191.4 (19.4)	545.0 (14.1)
REGULAR / unprimed	183.3 (14.4)	463.9 (8.8)	367.3 (12.8)	187.3 (17.5)	509.9 (11.6)

Offset



Error bars: ±1 SE.